UNIVERSITY OF TECHNOLOGY AND APPLIED SCIENCE – IBRI

MATHEMATICS FOR COMPUTING/MATHEMATIS FOR IT

MATH2101/MATH201. QUIZ – 1

$\frac{4}{4.5}$

Nan _____ ID _____

Choose the correct answer from the given. **[5x0.5=2.5Marks]**

1. Which among the following is a vector quantity?

   a) Mass b) Distance c) Speed d) Velocity

2. If matrix AB is of order 3×2 and matrix A is of order 3×3, then the order of B is

   a) 2×3 b) 3×2 c) 2×2 d) 3×3

   AB

   A B

   $3 \times 3 \quad 3 \times 2$

3. If $A = \begin{bmatrix} 0 & x+3 & x+2 \\ -4 & 3 & -4 \\ -5 & -4 & 2 \end{bmatrix}$ is a symmetric matrix, then $x=$

   a) -7 b) -1 c) 1 d) 7

   $x+3 = -4$ (3) $x+2 = -5$ (-7)

   $x = -7$ $x = -7$

4. Which among the following vectors is collinear?

   a) $\vec{u} = (1,4,2)$, $\vec{v} = (-1,4,-2)$ b) $\vec{u} = (5,-4,2)$, $\vec{v} = (-5,4,-2)$ c) $\vec{u} = (1,-4,-2)$, $\vec{v} = (-1,4,-2)$ d) $\vec{u} = (-1,4,2)$, $\vec{v} = (-1,4,-2)$

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5. A vector that is perpendicular to the vector $\vec{u} = (1,2,-1)$ is (-1, -2, 1)

Solve the questions with necessary steps.

Express $(2,3,-6)$ as a linear combination of vectors $(1,2,3), (3,5,1)$ *and* $(0,0,8)$. **[1.5 Marks]**

$2, 3, -6 = a(1, 2, 3) + b(3, 5, 1) + c(0, 0, 8)$

$= a + 3b + 0 = 2$

$2a + 5b + 0 = 3$

$3a + b + 8c = -6$

using Calculator:

$a = -1$

$b = 1$

$c = -\frac{1}{2}$

$\frac{1}{2}$

7. If $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, show that $A^2 = 2A$. **[1.5 Marks]**

$A^2 =$

$2A = \begin{bmatrix} 1\times 2 & -1\times 2 \\ -1\times 2 & 1\times 2 \end{bmatrix}$

$A^2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1\cdot 1 + -1\cdot -1 & 1\cdot -1 + -1\cdot 1 \\ -1\cdot 1 + 1\cdot -1 & -1\cdot -1 + 1\cdot 1 \end{bmatrix}$

$= \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$

$2A = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$

$A^2 = 2A$

8. For the vector $\vec{u} = (2,2,1)$ and $\vec{v} = (1,2,-4)$, find **[2 Marks]**

a. $\vec{u}.\vec{v} =$ $(2\cdot 1 + 2\cdot 2 + 1\cdot -4) = (2+4-4) = 2$

b. $|\vec{u}| =$ $\sqrt{(2)^2+(2)^2+(1)^2} = \sqrt{4+4+1} = \sqrt{9} = 3$

c. $|\vec{v}| =$ $\sqrt{(1)^2+(2)^2+(-4)^2} = \sqrt{1+4+16} = \sqrt{21}$

d. $\theta =$ _____, here θ, is the angle between the vectors.

$\cos\theta = \dfrac{u\cdot v}{\sqrt{\vec{u}}\,\sqrt{\vec{v}}} = \dfrac{2}{3\cdot\sqrt{21}}$

$= \cos^{-1}\theta = \left(\dfrac{2}{3\cdot\sqrt{21}}\right)$

$= 81.63$ by Degree

$= 1.42$ by Radian

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